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SUPERSEDING
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(See Section 6)

MILITARY SPECIFICATION

MOLDING COMPOUND, REVERSION RESISTANT, NON-CARCINOGEN CURING POLYURETHANE, AIRCRAFT STRUCTURE

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers the requirements for two component, chemically elevated temperature curing, reversion resistant polyether polyurethane compounds for filling and sealing aircraft structures for added strength and protection against the passage of air, liquids and vapors. Component materials and cured compound shall not contain any known carcinogens..

* 1.2 Classification. The molding compound shall be of the following types and classes as specified (see 6.2.1)

Type I - Two component
Type II - Premixed, degassed and frozen

Class 1 - Hardness point range of 65 ±5
Class 2 - Hardness point range of 80 ±5
Class 3 - Hardness point range of 90 ±5

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9320

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SPECIFICATIONS

FEDERAL

QQ-A-250/4	- Aluminum Alloy, 2024, Plate and Sheet
QQ-A-250/13	- Aluminum Alloy Alclad 7075, Plate and Sheet
PPP-B-585	- Box, Wood, Wirebound
PPP-B-601	- Box, Wood, Cleated - Plywood
PPP-B-621	- Box, Wood, Nailed and Lock-Corner
PPP-B-636	- Box Fiberboard
PPP-C-96	- Can, Metal, 28 Gage and Lighter

MILITARY

MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-S-18729	- Steel Plate, Sheet and Strip Alloy 4130, Aircraft Quality
MIL-L-23699	- Lubricating Oil, Aircraft Turbine Engines, Synthetic Base

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 395	- Compression Set of Vulcanized Rubber
ASTM D 412	- Tension Testing of Vulcanized Rubber
ASTM D 518	- Surface Cracking Resistance of Stretched Rubber Compounds
ASTM D 624	- Tear Resistance of Vulcanized Rubber
ASTM D 792	- Specific Gravity and Density of Plastics by Displacement
ASTM D 1149	- Accelerated Ozone Cracking of Vulcanized Rubber
ASTM D 2240	- Indentation Hardness of Rubber and Plastics by Means of a Durometer
ASTM D 2572	- Isocyanate Groups in Urethane Materials or Prepolymers
ASTM D 2849	- Urethane Foam Polyol Raw Materials

(Application for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

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CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 First article inspection. The molding compound furnished under this specification shall be a product which has been inspected and has passed the first article inspection specified herein (see 4.3).

* 3.2 Material. Type I compound shall be supplied as a two component system consisting of a polyether polyurethane prepolymer and a curing agent. Type II compound shall be supplied mixed, degassed and frozen. The properties of both types after curing are the same. The prepolymer shall cure through the addition of the curing agent and application of an external source of heat. This material is not dependent on solvents or moisture for curing.

* 3.2.1 Primer. The use of a primer to improve the adhesion of the compound to the substrate is permitted. The primer shall not be corrosive to the metal substrate or cause deterioration of other materials and shall be used in all tests where required for the compound to conform to specification requirements.

* 3.2.2 Appearance. Each component, and mixed compounds shall have a uniform homogeneous texture and shall be free of coarse particles, foreign matter and lumps. Storage of the materials shall cause no separation or crystallization of ingredients which cannot be readily re-dissolved on heating to 120°F.

* 3.3 Toxicity. Each component and mixed uncured compound, or cured compound shall have no adverse effect on the health of personnel engaged in the mixing and application of the systems under conditions of adequate ventilation and while exercising caution to avoid prolonged contact with the skin and while observing Occupational Safety and Health Administration (OSHA) guidelines (see 6.4).

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3.4 Color. Unless otherwise specified in the contract or order, the compound shall be furnished in the color manufactured.

3.5 Characteristics and performance.

3.5.1 Properties before curing.

3.5.1.1 Nonvolatile content. The nonvolatile content of the mixed compound shall be not less than 99 percent when tested as specified in 4.6.1.

3.5.1.2 Application life. The requirements for application life shall include the determination of the initial viscosity of the mixed Type I compounds, initial flow rate of Type II compounds when brought to room temperature, and the application time.

* 3.5.1.2.1 Initial viscosity of mixed Type I compound (all classes). The initial viscosity of the mixed Type I compounds shall be 8000 to 12000 centipoises.

3.5.1.2.2 Initial flow rate of Type II (all classes). The initial flow rate range of Type II compounds shall be from 25 to 550 grams per minute.

3.5.1.2.3 Application time. The application time of all mixed compounds shall be not less than 90 minutes (Minimum specified time to reach 250,000 centipoises) when tested as specified in 4.6.3.

* 3.5.1.3 Isocyanate content(prepolymer). When tested as specified in 4.6.18, the minimum isocyanate content of the prepolymer shall be 10 percent.

* 3.5.1.4 Water content (curing agent). When tested as specified in 4.6.19, the maximum percent water present in the curing agent shall not exceed 0.20 percent.

3.5.2 Properties after cure.

3.5.2.1 Physical properties. The physical properties of the cured compound shall conform to the requirements of Table I.

3.6 Storage life.

3.6.1 Type I. When mixed and cured as specified in 4.5.2 and 4.5.4 respectively, the molding compound shall conform to all

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the requirements of Table I after the components have been stored in the original unopened containers at standard conditions (4.5.1) for 180 days from the date the product is received by the procuring activity (see 4.6.20).

3.6.2 Type II. Cured Type II compound shall conform to all the requirements of Table I after being stored at -30 ±2° F (-34.4 ±1.1° C) for 60 days from the date the product is received by the procuring activity (see 4.6.20).

4. *QUALITY ASSURANCE PROVISIONS*

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection of the molding compound shall be classified as follows:

a. First article inspection (4.3).

b. Quality conformance inspection (4.4).

4.3 First article inspection. First article inspection shall include all the examinations and tests of this specification except the examinations of preparation for delivery. Unless otherwise specified, no material shall be submitted for acceptance under any contract or order until the first article samples prescribed in 4.3.1 have been subjected to first article inspection and pronounced satisfactory by the procuring activity. However, approval of the first article samples shall not relieve the supplier of his obligation to meet the quality conformance inspection (4.4). The responsibility for the performance of the first article inspection shall be specified by the procuring activity (see 6.2.1).

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4.3.1 First article test samples. Samples for first article inspection shall be as follows:

Type I - A minimum of 5 quarts of prepolymer together with required amount of curing agent.

Type II - A quantity of frozen cartridges approximately equal to the amount furnished as Type I.

In addition, when required, sufficient primer (see 3.2.1) for test purposes shall accompany the samples. Samples shall be selected at random from material which has been manufactured using the same production processes, procedures and equipment which will be used in filling the contract.

* 4.3.2 Data to accompany test samples. Two copies of the manufacturer's test report shall be submitted simultaneously with the samples of 4.3.1. The report shall contain numerical test data, where applicable, showing that the material submitted for the first article inspection conforms to the requirements of this specification. In addition, two copies of the manufacturer's instructions for the mixing and application of the compound shall accompany the report, and application of the primer, when required.

4.3.3 First article sample and inspection for a subsequent contract. If a contractor has previously furnished molding compound in accordance with the requirements of this specification and his product has been found to be satisfactory, the requirements for a first article sample and its submittal for any subsequent contract or order may be waived at the discretion of the procuring activity (see 6.3).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations specified in 4.4.2.2 through 4.4.2.4 and the tests specified in Table II for each lot of material submitted for acceptance.

4.4.1 Sampling.

4.4.1.1 Lot formation. Unless otherwise specified, a lot shall consist of all the material of the same Type and Class manufactured at one time from one batch (see 6.5), forming part of one contract or order and submitted for acceptance at the same time and place.

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4.4.1.2 Sampling for tests. A sufficient number of containers shall be selected at random to allow preparation of the specimens for each test in Table II. These containers shall be sealed, marked, and held for testing purposes.

4.4.1.3 Sampling for visual inspection of filled containers. A random sample of filled containers shall be selected in accordance with MIL-STD-105, Inspection Level I. Acceptable Quality Level (AQL) of 2.5 percent defective for examination as specified in 4.4.2.2 through 4.4.2.4.

4.4.1.4 Report of tests. The manufacturer shall submit test reports to the Government representative for each lot showing the results including numerical values when applicable of all the tests specified in Table II. Reports of numerical tests as satisfactory are not acceptable.

4.4.2 Inspection procedures.

4.4.2.1 Tests. The quality conformance tests shall consist of all the tests specified in Table II and such other tests as are necessary to check the manufacturer's data. If a test specimen fails to meet any of the tests required by this specification, the lot represented by the sample shall be rejected.

4.4.2.1.1 In addition, the Government reserves the right to have any of the tests called for by this specification rerun at a Government laboratory at any time within 180 days for Type I "as received" and 60 days for Type II material except as provided for storage life (see 3.6). Samples for these tests shall be taken from a previously unopened container of the submitted acceptance sample which shall have been stored as described in 4.6.20.

4.4.2.2 Examination of preparation for delivery. The samples selected in accordance with 4.4.1.3 shall be examined for packaging, packing, marking, leakage, and workmanship. Any package having one or more defects shall be rejected. If the number of defective packages exceeds the Acceptable Quality Level (AQL) of 2.5 percent defective in accordance with MIL-STD-105, the lot represented by the sample shall be rejected.

4.4.2.3 Examination for container fill. The sample size of 4.4.1.3 shall be adjusted to Inspection Level S-3 of MIL-STD-105 except that the sample size derived from MIL-STD-105 shall be rounded off to the next

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highest increment of five. At no time shall the sample size be less than five, unless the lot consists of less than five containers in which case all containers shall be used to determine the fill. The sample unit shall be one filled container. The lot shall be rejected if the average content per container is less than that specified.

4.4.2.4 Examination of product. Conformance of the material to the requirements for appearance and any other requirements not covered by specific tests shall be determined by appropriate visual examination.

4.5 Test conditions.

4.5.1 Standard conditions. Unless otherwise specified herein, all mixing, and testing shall be performed at 25 ±1.1°C (77 ±2°F) and a relative humidity of 50 ±5 percent.

* 4.5.2 Preparation of compound for tests.

4.5.2.1 Mixing of Type I compound. The base compound and its curing agent, both in the original unopened containers, together with the required spatulas, beakers, and other mixing equipment shall be held at standard conditions (4.5.1) for a minimum of 2 hours. The base compound and curing agent shall then be thoroughly mixed in the proportions recommended by the manufacturer. Proper care shall be taken to avoid incorporation of air by excessive stirring. The mixed material shall be degassed under vacuum.

4.5.2.2 Thawing of Type II compound. The premixed, frozen cartridges of Type II compound shall be thawed in a 130 ±5°F (54.4 ±1.1°C) water bath for 18 ±2 minutes and then placed at 77 ±2°F (25 ±1.1°C) for 18 ±2 minutes prior to testing.

4.5.3 Molding. Molds for disk specimens and special shapes as indicated in individual test descriptions shall be of polyethylene, polytetrafluoroethylene (PTFE) coated metal, or coated with any other material having a nonadhesive surface which will not transfer to the test specimen. The mold shall be constructed to facilitate removal of the specimen. The mold shall be thoroughly cleaned with a suitable solvent and dried prior to the application of the mold release agent and compound. The mixed compound shall be poured or extruded slowly into the mold. The mold shall then be covered with a flat, smooth plate and a pressure not to exceed 10 pounds per square inch (psi) applied.

* 4.5.4 Curing. Specimens prepared as specified in 4.5.2 through 4.5.3 shall be cured for 18 hours at 82.2 ±1.1°C (180 ±2°F). Cure time shall include the time specimens are in the molds. Cured specimens shall be kept at standard conditions at least 24 hours prior to testing unless otherwise specified herein.

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4.6 Test methods.

4.6.1 Nonvolatile content. Five to 10 grams of the mixed compound shall be transferred as soon after mixing as possible to a dish about 8 centimeters (cm) in diameter. A tight fitting cover shall immediately be placed over the dish and the weight determined to the nearest milligram. The cover shall then be removed and the compound heated for 24 hours at 82.2 ±5.5°C (180 ±10°F). It shall then be cooled in a closed desiccator, removed from the desiccator, and the weight determined to the nearest milligram. The percent nonvolatile content shall be calculated as follows:

$$\text{Percent Nonvolatile Content} = \frac{\text{Final Weight}}{\text{Initial Weight}} \times 100$$

4.6.2 Initial viscosity and flow rate.

* 4.6.2.1 Initial viscosity (Type I). A 250 gram sample of freshly mixed Type I compound shall be used to determine viscosity. Mix the sample for 3 minutes and determine its viscosity with a Brookfield viscometer model RVF (or equivalent) using a No. 4 spindle and speed of 10 RPM. Readings shall be recorded after the spindle has run for 5 seconds. Retain the sample for determining the application time (4.6.3.1).

4.6.2.2 Initial flow rate (Type II). The cartridge shall be thawed as specified in 4.5.2.2. The nozzle of the cartridge shall have a 0.125 ±0.005 inch opening and the application gun set for 110 psi back pressure. Material shall be extruded initially to fill the nozzle and to remove any trapped air. After a period of exactly one hour and again after 90 minutes, measured from the end of the thawing period, extrude the material for ten seconds in a tared container. Determine the flow rate in grams per minute by extruding the material for ten seconds and converting the weight to grams per minute.

* 4.6.3 Application time.

4.6.3.1 Type I. Determine the viscosity of the sample of 4.6.2.1 15 minutes after completion of mixing using a Brookfield viscometer model RVF (or equivalent) using a No. 6 spindle and speed of 4 RPM. Supplemental readings shall be made periodically until a reading of 250,000 centipoises is attained, which shall be considered as the end of the application time. The application time shall commence at the end of the mixing period. The spindle shall not be drawn from the material during the test. For quality conformance testing (Table II), the sample shall also be used to determine hardness in accordance with 4.6.4.

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4.6.3.2 Type II. Type II compound shall be thawed as specified in 4.5.2.2. Determine its viscosity with a Brookfield viscometer model RVF (or equivalent) using a No. 6 spindle and speed of 4 RPM by extruding a sufficient amount of material and mixing the sample for 3 minutes. The test procedure specified in 4.6.3.1 shall be followed until a reading of 250,000 centipoises is attained, which shall be considered as the end of the application time.

4.6.4 Hardness. Six specimens having 2 by 2 inch minimum face dimensions and a minimum thickness of 0.250 ±0.003 inch shall be prepared and cured as specified in 4.5.4. The hardness shall be determined with a Type A hardness durometer in accordance with ASTM D 2240 using a 3 second application time. Two specimens shall be subjected to the heat aging test (4.6.16), two specimens retained for the oil resistance test (4.6.15) and two specimens for determining hardness after hydrolytic exposure (4.6.17).

4.6.5 Tensile strength and ultimate elongation. Twelve specimens shall be cut from a sheet of the compound 0.075 ±0.005 inches thick, molded as specified in 4.5.3. The test specimens shall be cut using Die "C" and tested in accordance with ASTM D 412. Three specimens shall be saved for determining the tensile strength after heat aging (4.6.16) and three specimens for determining tensile strength and elongation after hydrolytic stability (4.6.17).

4.6.6 Shrinkage. A cubical mold, approximately 1.0 inch on each side and having an open top shall be constructed in accordance with 4.5.3. Its volume at standard conditions (4.5.1) shall be determined. A compound specimen shall be prepared in the mold and cured in accordance with 4.5.4. After curing, the specimen shall be placed in a circulating air oven at 82.2 ±2.7°(180 ±5°F) for 96 hours. It shall then be removed, cooled, and its volume at 25 ±1.1°C (77 ±2°F) determined by the water displacement method. The percent shrinkage shall be calculated as follows:

$$\text{Percent Shrinkage} = \frac{V_1 - V_2}{V_1} \times 100$$

where: V_1 = Volume of mold

V_2 = Final volume of compound

4.6.7 Specific gravity. Two test specimens of 4.6.4 shall be used for determining the specific gravity. Tests shall be in accordance with ASTM D 792.

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4.6.8 Tear strength. Three tear strength specimens shall be cut from a sheet of the compound 0.075 ±0.005 inches thick, molded as specified in 4.5.3. The test specimens shall be cut using Die "C" and tested in accordance with ASTM D 624.

* 4.6.9 Compression set. Three specimens 0.500 ±.005 inches thick and 1.129 ± 0.001 inches in diameter shall be molded in accordance with 4.5.3. The compression set shall be determined in accordance with Method B, compression set under constant deflection of ASTM D 395. The test shall be conducted at 70°C (158°F) for 22 hours.

4.6.10 Adhesion.

4.6.10.1 Adhesion to steel and aluminum panels.

* 4.6.10.1.1 Preparation of specimens. When required, the surface of the metal panels listed in Table III shall be vapor degreased, sand blasted, and solvent wiped with methyl-ethyl-ketone. To the cleaned steel and aluminum surfaces, apply a coat of primer (see 3.2.1) when required. Application, film thickness, drying time and required conditions shall be specified by the manufacturer. A sufficient quantity of mixed molding compound shall be applied to the panels to form a coating having a thickness of .125 inch ±0.030 inch. A 3 inch by 6 inch area of a 3 inch by 12 inch strip of 60 mesh corrosion resistant metal screen shall be intimately coated with the mixed compound and placed on the coated area of each panel leaving a loose end 6 inches long. The panels shall be cured in accordance with 4.5.4.

4.6.10.1.2 Immersion of specimens. Immediately after completion of curing, the test specimens shall be immersed vertically for 7 days ±4 hours in a closed glass vessel containing a 3 percent aqueous sodium chloride solution with 4 inches of the panel exposed to the test fluid and the remainder of the panel exposed to the air vapor mixture. The temperature during the test shall be maintained at 70 ±1.1°C(158°+2°F). At the expiration of the test period the fluid shall be cooled to 25 ±1.1°C (77° ±2°F). The panels shall then be removed and the adhesion determined in accordance with 4.6.10.1.3, within 10 minutes after removal from the fluid.

4.6.10.1.3 Testing of specimens. Specimens prepared and conditioned in accordance with 4.6.10.1.1 and 4.6.10.1.2 shall be tested as follows: Two 1-inch wide strips shall be cut lengthwise through the screen and compound to the panel surface and extended the full length of the loose

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end of the screen. The edges of the panel shall not be used as one edge of the test strip. The specimens shall be tested individually in an autographic testing machine with a capacity such that the tension at failure is not more than 85 percent and not less than 15 percent of the full scale load. The rate of separation of the jaws shall be 12 inches per minute and shall be uniform at all times. Specimens shall be mounted in the machine so that the loose end of the 1-inch wide screen strip will be folded 180 degrees as it is pulled from the panel. Each strip shall be pulled individually. The adhesion in pounds shall be automatically recorded on a chart as a continuous curve. The adhesion value shall be calculated by averaging the maximum and minimum forces required to separate the compound from the panel. If cohesive failures occur, the adhesion value shall be reported as greater than the measured value.

4.6.11 Moisture absorption. A specimen 1.0 inch by 1.0 inch by 1.0 inch shall be molded in accordance with 4.5.3. The specimen shall be weighed and immersed for 24 hours in a distilled water bath held at 93.3 ±2.7°C (200° ±5°F). The specimen shall be removed, blotted dry and weighed. The percent of water absorption shall be calculated as follows:

$$\text{Percent } H_2O \text{ Absorption} = \frac{\text{Final Weight} - \text{Initial Weight}}{\text{Initial Weight}} \times 100$$

* 4.6.12 Low temperature flexibility. Three test panels of aluminum alloy alclad conforming to QQ-A-250/13, measuring 1 inch by 6 inches by 0.032 inch, shall be coated on one side with the primer (see 3.2.1) when required, and allowed to dry. A quantity of molding compound shall be mixed and applied to the primed surfaces in a manner which will produce cured coatings with a thickness of 0.050 inch to 0.060 inch, leaving 1 inch at each end of the panels uncoated. Immediately after completion of curing, the test panels shall be immersed in a 3 percent aqueous sodium chloride solution for a period of 3 weeks (504 hours ±12 hours). The test specimens shall then be inserted in a flexibility jig as shown in Figures 1 and 2 so that the uncoated side will contact the contour block and the weight will contact only the uncoated end of the panel. The flexibility jig and test specimens shall be subjected to a conditioning temperature of -53.9 ±1.1°C(-65° ±2°F) for 6 hours. After the specified conditioning, the specimens shall be bent around the curved portion of the flexibility test jig by releasing the fastening hook. The test specimens shall be removed and examined for failure due to blistering, cracking, or loss of adhesion.

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4.6.13 Fungus resistance. Three specimens 4 by 4 by 0.125 inch shall be molded and cured as specified in 4.5.3 and 4.5.4 respectively. Tests shall be conducted in accordance with MIL-E-5272.

4.6.14 Ozone resistance. Test specimens shall be molded and cured as specified in 4.5.3 and 4.5.4 to meet Table IV requirements. Test equipment shall be in accordance with ASTM D 1149. Test procedures shall be in accordance with Table IV.

4.6.15 Oil resistance. Two specimens from 4.6.4 shall be weighed to the nearest milligram and immersed in lubricating oil conforming to MIL-L-23699 for a period of 24 hours ± 10 minutes at 93.3 ±1.1°C (200 ±2°F). The oil bath containing the specimens shall then be cooled to standard conditions (4.5.1). Determine the hardness and change in weight of the specimens after oil immersion for conformance to Table I.

4.6.16 Heat aging. Test specimens prepared as specified in 4.6.4 and 4.6.5 shall be exposed in an air circulating oven for 168 hours at 100 ±2°C (212 ±5°F). The specimens shall then be removed and cooled for one hour at standard conditions (4.5.1). Determine the change in hardness (4.6.4) and tensile strength (4.6.5), after heat-aging for conformance to Table I.

4.6.17 Hydrolytic stability.

4.6.17.1 Specimens. The specimens prepared in 4.6.5 shall be used for the determination of the tensile strength after hydrolytic exposure. Hardness shall be determined on the specimens of 4.6.4 before and after hydrolytic exposure at the same location on the specimens using a 3 second application time.

4.6.17.2 Procedure. After determining hardness before exposure, the specimen shall be placed vertically in a suitable holder on a tray in a suitable glass desiccator (250 mm.). The desiccator shall contain a glycerine (22 percent by weight) in water solution in the bottom which will produce a minimum relative humidity (RH) of 95 percent at the test temperature. The desiccator, containing the specimens, shall then be closed and inserted into an air circulating oven maintained at 93.3 ±1.1°C (200 ±2°F) for a period of 28 days. At the end of the exposure period, the desiccator shall be removed from the oven and cooled to standard conditions (4.5.1) for 16 to 24 hours. The averaged results for the hardness, tensile strength, and ultimate elongation shall be recorded and checked for conformance to Table I.

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* 4.6.18 Isocyanate content (prepolymer). Isocyanate content shall be determined using ASTM D 2572.

* 4.6.19 Water content (curing agent). Curing agent water content shall be determined in accordance with ASTM D 2849, sections 60 to 69. Automatic instrumentation (see 6.6) may be used.

4.6.20 Storage life.

4.6.20.1 Type I. A minimum of one quart of base compound and the appropriate amount of curing agent, both in original, unopened containers, shall be stored at standard conditions (4.5.1) for 180 days. The compound shall be evaluated for conformance to 3.6.1.

4.6.20.2 Type II. Premixed frozen cartridges shall be stored for 60 days at -30 ±2°F (-34.4 ±1°C). After 60 days, the cartridges shall be thawed (4.5.2.2) and the material checked for conformance to 3.6.2.

5. PACKAGING

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Type I compound.

5.1.1.1.1 Unit protection. The base compound and curing agent shall be packaged in their own containers. The curing agent shall be packaged in such a way as to allow easy separation for mixing purposes but will prevent accidental separation. The ratio of the quantity contained in the attached curing agent container shall be the same as the recommended mixing ratio of the base compound and curing agent.

5.1.1.1.2 Unit package. The base compound shall be furnished in 1/2 pint, 1 pint, 1 quart, or 1 gallon multiple friction top cans, conforming to Type V, Class 2 of PPP-C-96. The amount of base compound contained in each can shall fill the container to approximately three-fourths of its capacity. The proper amount of curing agent shall be furnished in glass or plastic jars having vertical smooth inside walls, with projections or lips not exceeding 1/16 inch. The inside bottom shall be approximately horizontal. The jars shall have one rolled thread such that the cap will turn a minimum of 1/4 revolution to fully secure. The jars shall be provided with enameled metal or plastic screw type caps.

5.1.1.2 Type II compound. Type II compound shall be supplied premixed, degassed and quick frozen in 6 and 12 ounce plastic cartridges complete with plunger and cartridge cap or plug, as manufactured by Pyles Industries, Inc. and Semco Research Co., or equal as specified in the contract or order (see 6.2).

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5.1.2 Level C. Type I base compound with curing agent, and Type II compound shall be packaged in accordance with the manufacturer's commercial practice.

5.1.3 Instructions for use. The supplier of the molding compound shall furnish with each unit package printed label instructions which shall contain:

a. Storage life, uncatalyzed, indicating the optimum storage temperature and conditions of storage to obtain maximum storage life.

b. The proportions of compounds and curing agents of Type I material shall be on a weight basis and the methods for mixing them shall be clearly stated.

c. Application of primer, when required.

5.2 Packing. Packing shall be Level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Type I compound. The base compound and curing agent, packaged in accordance with 5.1.1.1, shall be packed in an overseas exterior type container conforming to PPP-B-585, PPP-B-601, PPP-B-621 or PPP-B-636. Weight of the contents shall conform to the weight limitations of the applicable box specification and shall not exceed a total weight of 200 pounds. Containers shall be closed and strapped in accordance with Appendix to PPP-C-96.

5.2.1.2 Type II compound. Type II unit packages as specified in 5.1.1.2 and frozen for shipment, shall be packed in a manner that will prevent thawing or other damage during transit. Shipping containers shall contain the same number of the type of unit packages, be of uniform size, and shall be designed to insure that damage is prevented during handling and shipping.

5.2.2 Level C. Types I and II material packaged in accordance with 5.1.2 shall be packed to afford protection against damage during direct shipment from the source of supply to the first receiving activity for immediate use. Containers shall comply to the Uniform Freight Classification Rules or other regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and exterior

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shipping container shall be marked in accordance with MIL-STD-129. Marking shall include specification number, type, and name of manufacturer; date of manufacture and expected life expiration date (month and year for Type I material and month, day, and year for Type II material) and recommended storage temperature.

6. NOTES

6.1 Intended use. The molding compound covered by this specification, in its cured condition, is intended for sealing and filling aircraft structures such as outboard ends of spars, rotor blade tips and surfaces.

6.2 Ordering data.

* 6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type and class of molding compound required (see 1.2).

c. Quantity desired.

d. Size of container.

e. Level of packaging and packing required (see 5.1 and 5.2).

f. Any special marking required.

g. Additional requirements, if applicable (e.g., primer see 3.2.1).

h. First article, when required (see 4.3.1 and 6.3).

* 6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements

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identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.3	First Article Inspection Reports	DI-T-5239-Inspection Test Reports

(Copies of data item descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Contracts or orders shall specify the following provisions for the first article inspection:

6.3.1 Whether first article inspection is required. When a contractor is in continuous production of the molding compound from contract to contract, consideration should be given to waive the first article inspections. If inspection is required, indicate:

a. Where the first article inspection is to be conducted (at the contractor's plant or Government or commercial laboratory).

b. That the approval of first article samples or the waiving of the first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6.4 Toxicity. Questions pertinent to the requirements of 3.3 should be referred, by the procuring activity, to the Navy Environmental Health Center, 3333 Vine Street, Cincinnati, Ohio 45220, who will act as an advisor to the procuring activity.

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6.5 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

* 6.6 Source for water content automatic instrument. An instrument recommended for use in 4.6.19, as a preferred alternate to those specified in ASTM D 2849, is the "Aquatest II" manufactured by Photovolt, New York, NY 10010.

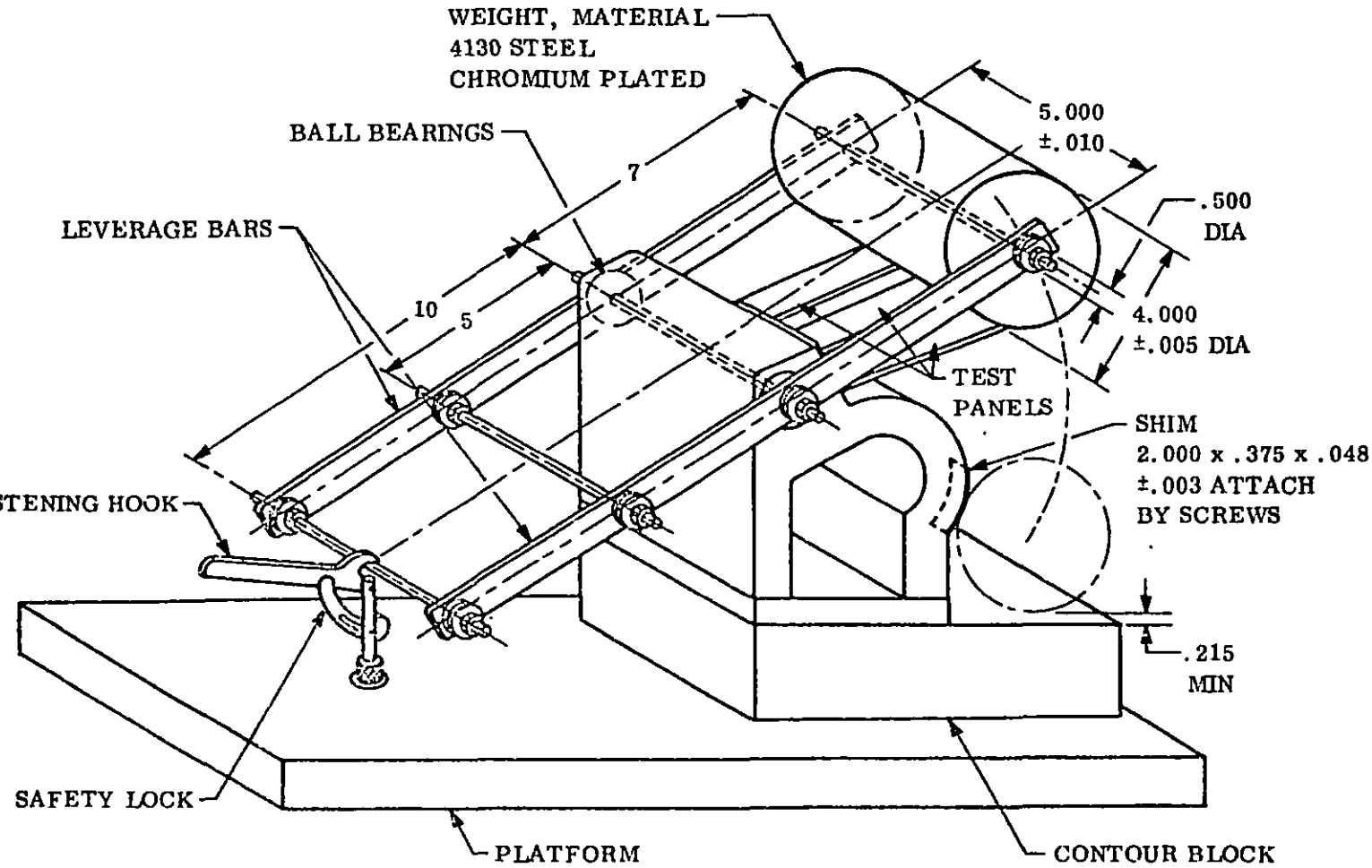
6.7 Supersedure data. This specification, MIL-M-81999A(AS), supersedes MIL-M-81999(AS) dated 7 July 1975. Molding compound covered in the superseded issue is available under this specification as follows:

MIL-M-81999(AS) Designation	MIL-M-81999A(AS) Designation
Type I	Type I, Class 1
	Type I, Class 2 (new)
	Type I, Class 3 (new)
Type II	Type II, Class 1
	Type II, Class 2 (new)
	Type II, Class 3 (new)

6.8 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - AS
(Project No. 9320-N257)

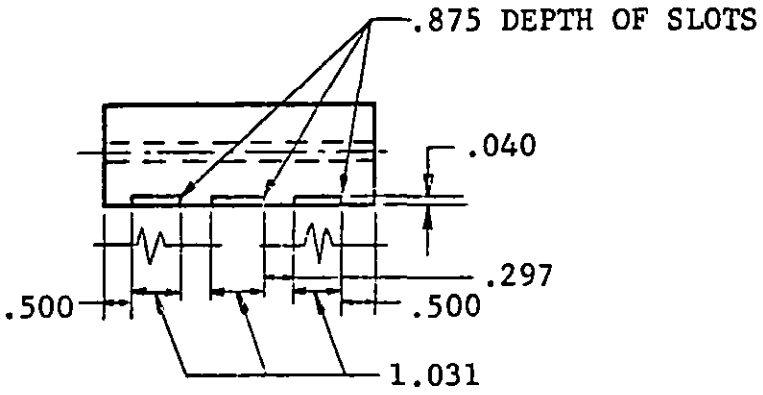
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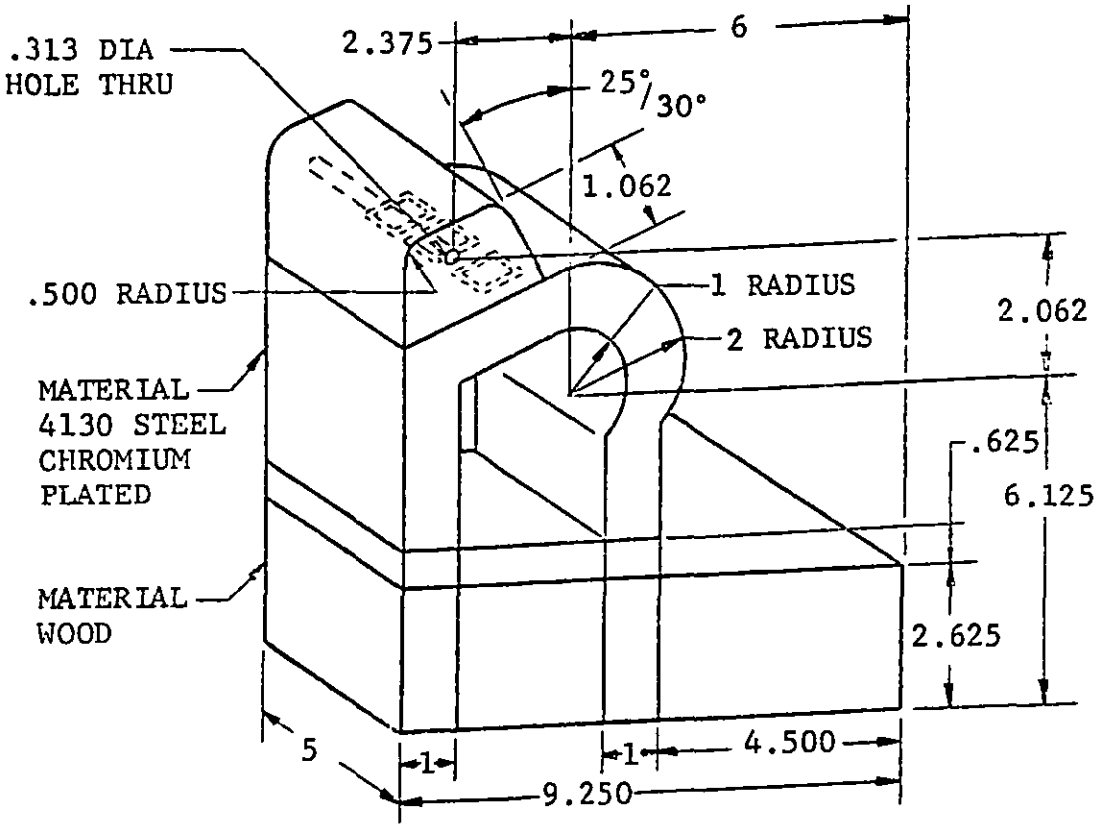
DIMENSIONS IN INCHES

FIGURE 1. Low temperature flexibility apparatus.
(see 4.6.12)

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VIEW SHOWING SLOTS FOR PANELS



DIMENSIONS IN INCHES.

FIGURE 2. Contour block.
(see 4.6.12)

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* TABLE I. Physical properties of cured compounds.
(see 3.5.2.1)

Physical property	Requirements			Test paragraph
	Class 1	Class 2	Class 3	
Hardness, Type A Durometer 1/	65 ±5	80±5	90 ±5	4.6.4
Tensile Strength, PSI, minimum	1600	3300	4000	4.6.5
Ultimate elongation, percent, min.	400	400	400	4.6.5
Shrinkage percent, maximum 4/	5.0	5.0	5.0	4.6.6
Specific gravity, range	1.0-1.2	1.0-1.2	1.0-1.2	4.6.7
Tear strength, ppi, min.	130	200	340	4.6.8
Compression set, percent max.	35	50	65	4.6.9
Adhesion, lbs/inch of width, min.				
To steel	30	40	50	4.6.10.1
To aluminum	30	40	50	4.6.10.1
Moisture absorption, percent max.	3.0	3.0	3.0	4.6.11
Low temperature flexibility at -53.9°C (-65°F)2/	Pass	Pass	Pass	4.6.12
Fungus resistance	Non-nutrient	Non-nutrient	Non-nutrient	4.6.13
Ozone resistance, 28 days exposure 3/	Pass	Pass	Pass	4.6.14
Oil resistance:				
Change in hardness, percent max.	40	40	40	4.6.15
Change in weight, percent max.	40	30	30	4.6.15
Heat aging				
Change in hardness, percent max.	15	15	15	4.6.16
Tensile strength after heat aging, PSI, min.	1500	3200	3500	4.6.16
Hydrolytic stability:				
Change in hardness points, max.	15	25	25	4.6.17
Tensile strength after exposure, PSI, min	1400	2300	3000	4.6.17
Elongation after hydrolytic stability, percent min.	300	300	300	4.6.17

1/ Test specimens shall be cured as specified in 4.5.4.

2/ There shall be no evidence of cracking, blistering, or loss or adhesion at -53.9±1.1°C (-65 ±2°F).

3/ There shall be no evidence of cracking as a result of ozone exposure when viewed under 7X magnification.

4/ Shrinkage values are based on cure cycles established in the specification. Changes in the cure cycle (lower temperature, etc) can yield lower shrinkage values.

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TABLE II. Quality conformance tests.
(see 4.4.2.1)

Requirement	Requirement paragraph	Test paragraph
* Before cure		
Initial Viscosity	3.5.1.2.1	4.6.2.1
Isocyanate content	3.5.1.3	4.6.18
Water content	3.5.1.4	4.6.19
Cured polymer		
Hardness	Table I	4.6.4
Tensile strength	Table I	4.6.5
Ultimate elongation	Table I	4.6.5
Specific gravity	Table I	4.6.7

TABLE III. Panels for adhesion test.
(see 4.6.10.1.1)

Composition	Specification	Size (inches)	Number of specimens cut from each panel assembly
Steel, Alloy 4130	MIL-S-18729	3 x 6 x 1/16	2
Aluminum Alloy 2024	QQ-A-250/4	3 x 6 x 1/16	2

TABLE IV. Ozone resistance specimens and procedure.
(see 4.6.14)

Test variable	Requirement
Test specimen 1/	ASTM D 518, Procedure B
Ozone concentration	50 ± pphm (volume)
Exposure period	28 days
Exposure temperature	40 ±1° C (104 ±2°F)
Magnification	7X

1/ Two specimens required.

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